

What is the Relationship between the Length (0.20m, 0.30m, 0.40m, 0.50m, 0.60m, 0.70m, 0.80m) and Period of Motion of a Conical Pendulum Rotating in Uniform Circular Motion at a Constant Angle?

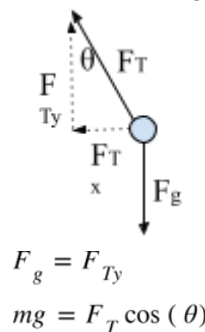
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Mr. Herbert's Physics 20/30IB  
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**Introduction:** Wave swinger rides such as the “Dream Machine” at Calaway Park always interested me.<sup>1</sup> Although I have a fear of some other rides, this one never scared me and I always wanted to keep riding it over and over again. This topic sparked my interest when we were introduced to pendulums in class. We conducted multiple simple pendulum experiments in class, but did we ever stop to consider other, more complex pendulums? The conical pendulum soon became my lab idea because of its interconnectivity with wave swinger rides, where floorless chairs are attached to a fixed top by chains. The spinning motion of the ride then causes the chairs to swing outward. I thought it would be interesting to investigate the effect of length on the period of a pendulum, thereby affecting the speed at which the pendulum rotates at.

**Background Information:** In this investigation, the length of a conical pendulum is altered where mass and angle of release remains the same. The period of a conical pendulum is potentially affected by the angle of the pendulum, mass of the bob, and length of the string; all factors which affect the speed of rotation. We can start off by considering the period formula for a simple pendulum:

$$T = 2\pi\sqrt{\frac{\ell}{g}}$$

Where T is the period of the pendulum,  $\ell$  is the length of the pendulum, and g is the gravitational field strength on Earth, which is approximately  $9.81\text{m/s}^2$ . This formula only works for small angles of release; surprisingly, mass and amplitude do not change the period of a simple pendulum. Here is the Free Body Diagram for the pendulum bob where m is mass of bob in kg and v is velocity in m/s.



**Equation 1.**

Equation 1 comes from analyzing the vertical components. The angle labeled  $\theta$  in the diagram is kept constant at  $20^\circ$ . It is clear that the vertical component of the tension force in the string balances out the force of gravity. This leaves the horizontal tension component behind. This horizontal force is equal to the centripetal force. An equation can be made as shown:

$$F_{Tx} = F_c$$

$$F_T \sin(\theta) = \frac{mv^2}{r}$$

We can replace velocity with angular velocity and radius using the equation:

$$\omega = \frac{v}{r} \quad v = \omega r$$

$$F_T \sin(\theta) = m\omega^2 r$$

**Equation 2.**

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<sup>1</sup> Unknown (n.d.) <Wave Swinger | Classic Swing Ride | Cedar Point> Reference to Calaway Park: Dream Machine.

The period can be found by then combining **Equation 1** and **Equation 2**:

$$\frac{F_T \sin(\theta)}{F_T \cos(\theta)} = \frac{m\omega^2 r}{mg}$$

$$\tan(\theta) = \frac{\omega^2 r}{g}$$

$$\omega = \sqrt{\frac{g \times \tan(\theta)}{r}}$$

$$\sin(\theta) = \frac{r}{\ell}$$

Replace  $r$  with  $\ell \times \sin(\theta)$  and  $\omega$  with  $\frac{2\pi}{T}$

$$\frac{2\pi}{T} = \sqrt{\frac{g \tan(\theta)}{\ell \sin(\theta)}}$$

$$T = 2\pi \sqrt{\frac{\ell \cos(\theta)}{g}}$$

### Equation 3.<sup>2</sup>

This is Equation 3, and is derived for the period of a conical pendulum where  $T$  is the period of the pendulum,  $\ell$  is the length of the pendulum (at an angle  $\theta$  from the normal, and  $g$  is acceleration due to gravity.<sup>3</sup> Similar to how a simple pendulum can demonstrate simple harmonic motion, a conical pendulum can demonstrate circular motion. The forces on a simple pendulum are different from that of a conical pendulum because a conical one will have centripetal force. As well, a conical pendulum's bob travels approximately twice the distance than a simple pendulum does which poses air resistance as a significant factor that affects results.

### Hypothesis:

Based on the research above, if the length of a conical pendulum is increased, then the period of the

pendulum will also increase. In the formula  $T = 2\pi \sqrt{\frac{\ell \cos \theta}{g}}$ , period is proportional to the square root of length. If length increases, period will also increase but not as drastically compared to the amount you change the length. When the length is small, the period should be small and friction would have a greater percentage of effect as compared to a longer period. Conceptually, this makes sense because a conical pendulum is quite similar to a simple pendulum which also follows this relationship between length and period.

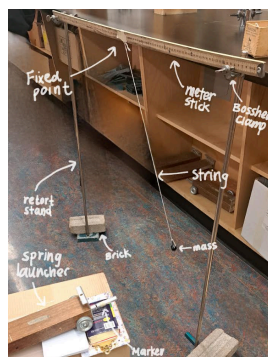
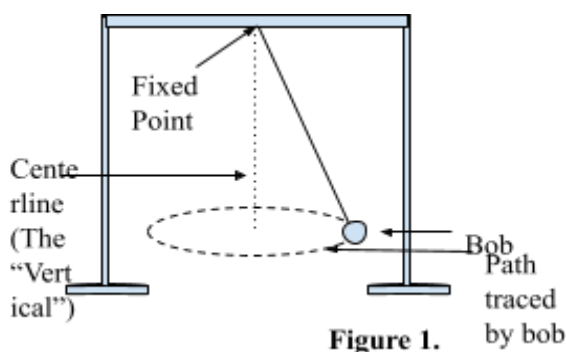
### Experimental Design:

A conical pendulum will be created using two large retort stands and a meter stick, the mass will be put into motion using a spring launcher. The length of the pendulum will be varied 7 times and the time for 5 rotations will be measured, 5 times each using a stopwatch.

<sup>2</sup> More, Hemant. (2020). *Conical Pendulum*. Retrieved February 16, 2024 from *The Fact Factor* <Conical Pendulum: Its time period, tension in string, numerical problems (thefactfactor.com)>

<sup>3</sup> Huygens, Christian. (1673). *Horologium Oscillatorium*. Translated by Bruce, Ian (2007) Retrieved January 28, 2024 from 17 Century Maths <17centurymaths.com/contents/huygenscontents.html>

**Diagram:** \*Diagram and photo are my own, produced using Google Drawings.



### Manipulated Variable:

Length of the pendulum string: Measured in meters using a meter stick from the fixed point of the pendulum to the point of attachment on the bob. The lengths used are 0.20, 0.30, 0.40, 0.50, 0.60, 0.70, and 0.80m. The uncertainty for the length will be  $\pm 0.001\text{m}$ , the smallest increment on the meter stick.

### Responding Variable:

The time taken for the pendulum to complete a full rotation measured in seconds (period) using a Samsung A53 Phone's stopwatch. The stopwatch will be started the moment the bob returns to the starting position after being launched, and stopped after 5 revolutions. The uncertainty for this measurement is  $\pm 0.01\text{s}$ . Divide by 5 to find the period.

### Controlled Variables:

1. **Mass of the Pendulum Bob:** The same mass of 50g was used for my experiment and was measured with a digital mass scale for precision. The result I have obtained is  $50.16\text{g} \pm 0.01\text{g}$ .
2. **Angle of Release and Angle of Rotation:**  $20^\circ$ , measured using a large protractor. I personally dragged the mass to the angle and held the string taut. Then I adjusted the lower support (books) so that the bob would stay in that position. I made sure the spring launcher provided enough force to release the bob and keep it spinning in a cone which forms a 20 degree angle with the vertical. Otherwise, the angle of rotation is difficult to control and this experiment acts on the assumption that the angle of release is identical to the angle it rotates at.
3. **Force applied to start the spinning:** Use the same spring launcher to launch the bob each trial. Push the tip of the spring launcher as far as the tape marker ( $\approx 10\text{cm}$  from the tip). Then place the tip of the launcher right next to the mass and let go quickly.
4. **Environmental Conditions:** Make sure there isn't wind or any factors that may affect things such as gravity. This is important as there may be lots of other people working in the lab. Make sure no fans or heaters are set up close by. In the long run, this is not a factor that would affect my results too much.
5. **Method of Measurement:** Having the same person perform the period measurements so that reaction time is kept constant. Measure the time it takes for the pendulum to make 5 rotations and divide by the number of revolutions to find the period. I personally timed each trial keeping reaction time constant. I start the stopwatch as the mass comes back to its start point and measure for 5 rotations. This would be divided by 5 in the analysis in order to obtain the period.

**Materials:**

1. 7 lengths of thin cotton string (0.20m, 0.30m, 0.40m, 0.50m, 0.60m, 0.70m, and 0.80m).
2. A 50g Mass with hook
3. 2 Bosshead Clamps
4. **2 Large Retort Stands**  
≈1.00m tall.
5. 2 Meter Sticks
6. Samsung Galaxy A53 Built in Stopwatch
7. Protractor
8. Scissors
9. Masking Tape
10. Digital Mass Scale/Balance
11. Books of Various Thicknesses
12. Bricks

**Procedure:**

1. Set up two large retort stands approximately a meter away from each other. Use a bosshead clamp on each retort stand, clamping them in at the same height near the top of each stand.
2. Horizontally fasten a meter stick to the clamps so that the meter stick fits snugly. If the retort stands aren't stable, weigh the base down using bricks.
3. Measure mass of the bob on electronic scale and record.
4. Cut out the strings. Make sure to make the strings a little longer so you can tie one end to the meter stick and the other to the hook of the mass. Cut off the excess afterwards.
5. Tie one end of the first length of string to the mass which acts as the bob of the pendulum. Tie the other end of the string to the center of the horizontal meter stick making sure the bar is high enough that the mass does not touch the ground.
6. Use tape to fix the end of the string on the horizontal bar. Making sure that the string will stay in place when lightly pulled on.
7. Pull the string outwards until the angle formed between the vertical (the string in rest position) and the string is  $20^\circ$  and place a protractor upside down parallel to the horizontal meter stick.
8. Support the bob from the bottom using a cardboard box while making sure the angle from Step 4 is constant. Adjust the height of the structure so it can be balanced. See Figure 3 Below.
9. With the help of a protractor, place a spring launcher perpendicular ( $90^\circ$ ) to the line joining the center of rotation of the mass and the mass, right in front of the mass. See Figure 3 Below.
10. Launch the mass using the spring launcher, it should trace out a conical shape. Push the spring all the way in and move it so the tip is next to the mass, then launch.
11. As soon as you launch the mass, remove the support so it doesn't get in the way.
12. When the mass comes back to the start position, start the stopwatch. Stop the stopwatch when 5 complete cycles have passed. Divide this time by 5 and record.
13. Repeat the measurement five times for each of the seven lengths.
14. Repeat Steps 7-13 with the six other lengths of string and record data.

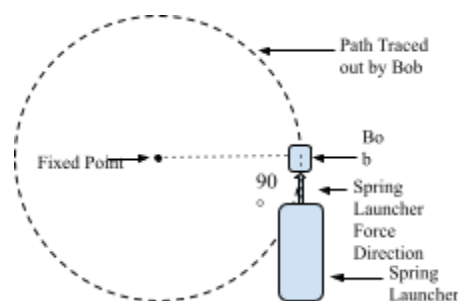


Figure 3: Birds Eye view of the pendulum.

**\*\*Environmental and Safety Considerations:** When operating the mass launcher, make sure there is enough space on all sides of the pendulum setup so that the bob won't accidentally hit anyone.

**Observations:**

1. Mass of bob:  $50.16\text{g} \pm 0.01\text{g}$
2. The angle of release is assumed to be in accordance with the angle of its spin. That is, when the bob is released at  $20^\circ$ , the angle formed between the vertical and the string stays constant at  $20^\circ$ .
3. Gravity is assumed to be  $9.81\text{m/s}^2$  even though it differs with different land altitudes where the experiment is performed.
4. The length measured when cutting the string is assumed to be accurate even though I see that the mass weighs down the string, stretching it a bit when attached to the pendulum.

**Table 1:** Period of different lengths of a Conical Pendulum released at  $20^\circ$  using a spring launcher.

| Length<br>$\pm 0.001\text{m}$ | Period $\pm 0.01\text{s}$ |         |         |         |         | Average of 5<br>Trials (s) | Absolute Period<br>Uncertainty (s) |
|-------------------------------|---------------------------|---------|---------|---------|---------|----------------------------|------------------------------------|
|                               | Trial 1                   | Trial 2 | Trial 3 | Trial 4 | Trial 5 |                            |                                    |
| 0.200                         | 0.98                      | 0.94    | 0.96    | 0.98    | 0.97    | 0.97                       | 0.03                               |
| 0.300                         | 1.10                      | 1.10    | 1.14    | 1.13    | 1.12    | 1.12                       | 0.03                               |
| 0.400                         | 1.26                      | 1.28    | 1.27    | 1.28    | 1.26    | 1.27                       | 0.03                               |
| 0.500                         | 1.39                      | 1.44    | 1.46    | 1.40    | 1.43    | 1.42                       | 0.03                               |
| 0.600                         | 1.53                      | 1.57    | 1.54    | 1.60    | 1.53    | 1.56                       | 0.03                               |
| 0.700                         | 1.63                      | 1.64    | 1.64    | 1.65    | 1.64    | 1.64                       | 0.03                               |
| 0.800                         | 1.72                      | 1.74    | 1.68    | 1.69    | 1.74    | 1.71                       | 0.03                               |

\*Note: All data has been rounded to 2 decimal places. For calculations and graphing, unrounded values were used for precision.

**Qualitative Observations:**

5. Angle of Release is kept constant at an angle of  $20^\circ \pm 1^\circ$ . This angle is seen to decrease as more time passes, however in the time span of five revolutions, this decrease seems to be quite small.
6. The first revolution is at a higher speed than the others because the bob has just been released by the spring launcher - This is why the first revolution is not counted for period.
7. The bob does not always trace out a circular path every time. Often, it's an elliptical shape.

**Sample Calculation for Average Period:** To find the average time out of 5 trials, take the sum of all five trials and divide by the number of trials which is 5.

$$\bar{x} = \frac{0.98\text{s} + 0.94\text{s} + 0.96\text{s} + 0.98\text{s} + 0.97\text{s}}{5} = 0.97\text{s}$$

### Sample Calculation for Average Data Value Uncertainty: Standard Deviation

To find the uncertainty for each data value, The formula below is used where  $\bar{X}$  is the mean,  $X$  is the individual data values and  $N$  is the number of trials.

$$\sigma = \sqrt{\frac{\sum (X - \bar{X})^2}{N}} = \sqrt{\frac{(0.98 - 0.97)^2 + (0.94 - 0.97)^2 + (0.96 - 0.97)^2 + (0.98 - 0.97)^2 + (0.97 - 0.97)^2}{5}}$$

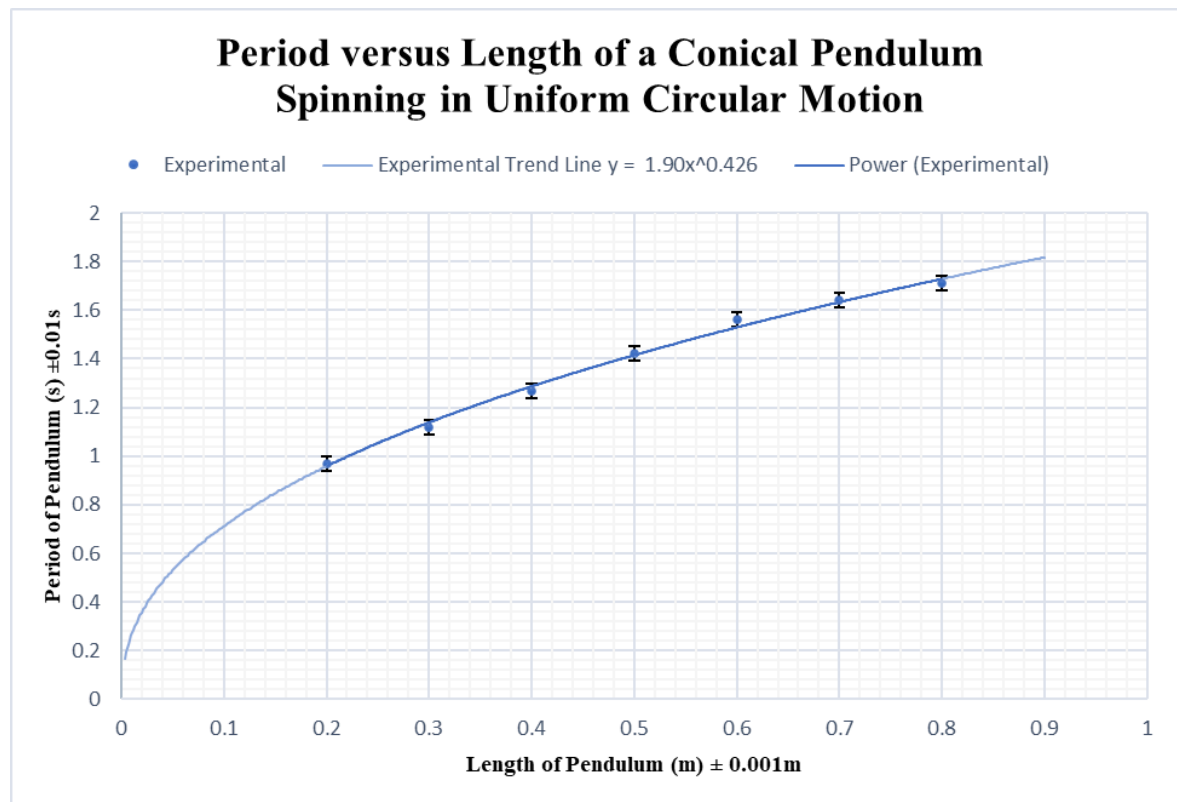
$\approx 0.0155s$

Therefore the repeated trial uncertainty for the length of 0.200m is  $\pm 0.02m$

### Uncertainties:

The instrument uncertainty for the stopwatch in the measuring period is  $\pm 0.01s$  and the repeated trials uncertainty is calculated as shown above for each length. The overall repeated trials uncertainty is significantly greater than the instrument uncertainty and will therefore be used as the final uncertainty. The highest uncertainty obtained through standard deviation is  $\pm 0.03s$  and will be the uncertainty for all values because reaction time and friction affect my confidence with the time obtained experimentally.

**Graph 1:** Graph of Period versus Length of a Conical Pendulum with data obtained Experimentally.



*\*Horizontal (length) uncertainty is small (0.001m) and is not shown on the graph above. I am confident that the error from the measurement of the string is minimal as a meter stick with scales measuring as small as a millimeter was used. Therefore the error is negligible.*

From this graph, we can see a square root relationship between period and length of a pendulum. This matches the hypothesis stating that period is proportional to the square root of length. As length increases, the period of the pendulum also increases, but at an increasingly slower rate of increase. The graph is seen to go through every point's small error bars which shows the relatively high precision of the data.

**Table 2:** Curve Straightening Values and Uncertainty for the Effect of Length on the Period of a Conical Pendulum.

| <b>Period (s)</b> | <b>Absolute uncertainty for Period (s)</b> | <b>Square of Period (s)</b> | <b>Percent Uncertainty (%)</b> | <b>Absolute Uncertainty (s)</b> |
|-------------------|--------------------------------------------|-----------------------------|--------------------------------|---------------------------------|
| 0.97              | 0.03                                       | 0.94                        | 6.19                           | 0.06                            |
| 1.12              | 0.03                                       | 1.25                        | 5.36                           | 0.07                            |
| 1.27              | 0.03                                       | 1.61                        | 4.72                           | 0.08                            |
| 1.42              | 0.03                                       | 2.02                        | 4.23                           | 0.09                            |
| 1.56              | 0.03                                       | 2.43                        | 3.85                           | 0.09                            |
| 1.64              | 0.03                                       | 2.69                        | 3.66                           | 0.10                            |
| 1.71              | 0.03                                       | 2.92                        | 3.51                           | 0.10                            |

### Sample Calculation for Uncertainty on the Curve Straightened Graph:

Length =  $0.200 \pm 0.001\text{m}$  and Experimental Period is  $0.97\text{s}$

$$(0.97 \pm 0.03\text{s})^2 = 0.94 \pm 2\left(\frac{0.03}{0.97} \cdot 100\%\right) = 0.94 \pm 6.19\%$$

$$0.06185... \cdot 0.94 \text{ (Square of Period)} = 0.0581... = 0.06\text{s}$$

$$= 0.94\text{s} \pm 6.19\% \text{ or } 0.94\text{s} \pm 0.06\text{s}$$

The original formula derived above can be manipulated to find what slope is equal to:

$$\text{Slope} = \frac{\text{Rise}}{\text{Run}} = \frac{\Delta y}{\Delta x} = \frac{T^2}{\ell}$$

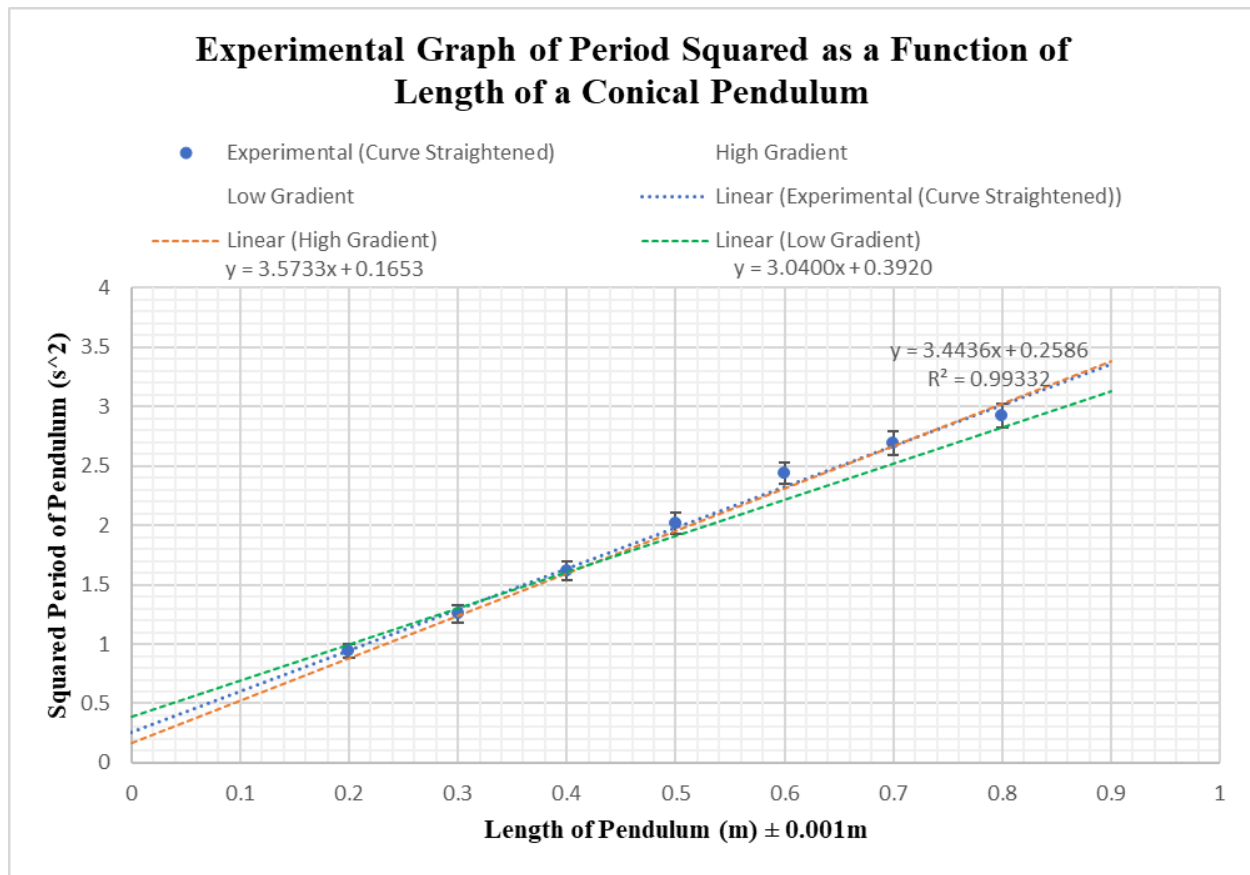
$$T = 2\pi\sqrt{\frac{\ell \cos 20}{9.81}}$$

$$T^2 = \frac{4\pi^2 \ell \cos 20}{9.81}$$

$$\frac{T^2}{\ell} = \frac{4\pi^2 \cos 20}{9.81}$$

**Graph 2:** Curve Straightened Graph for Period as a function of length of a Conical Pendulum.

\*Horizontal Error Bars are very minimal and are not shown in the graph. This error does not pose a significant change in results as I am confident that length is accurate and precise.



From the shape of the graph, it is evident that the experimental data follows a similar trend to the theoretical trend. However, the placement of the experimental data is slightly higher than theoretically predicted and not all values go through the error bars. This is a systematic error and weakens the accuracy of the data.

The trend line is defined by the equation  $y = 3.4436x + 0.2586$ . The slope of the trend line is 3.4436, and the slope of the high and low gradients respectively is 3.5733 and 3.0400.

$$|low\ gradient\ slope - trend\ line\ slope| = \left| \frac{3.0400s^2}{m} - \frac{3.4436s^2}{m} \right| = \frac{0.4036s^2}{m}$$

$$|high\ gradient\ slope - trend\ line\ slope| = \left| \frac{3.5733s^2}{m} - \frac{3.4436s^2}{m} \right| = \frac{0.1297s^2}{m}$$

The difference between the low gradient and the slope of the trend line is greater and will therefore be the one used as the uncertainty. The final slope ( $\frac{T^2}{\ell} = \frac{4\pi^2 \cos 20}{9.81}$ ) is  $3.4436 \pm 0.4036s^2/m$ .

#### Percent Uncertainty:

$$Percent\ Uncertainty = \left| \frac{Absolute\ uncertainty}{Data\ value} \right| \cdot 100\%$$

$$= \left| \frac{0.4036s^2/m}{3.4436s^2/m} \right| \cdot 100\% = 11.72029...\% = 11.72\%$$

To compare this slope with a theoretical value, the theoretical period can be found by substituting lengths into Equation 3. No uncertainties are included because theoretically all values are exact.

**Table 3:** Theoretical Period Values on a Conical Pendulum

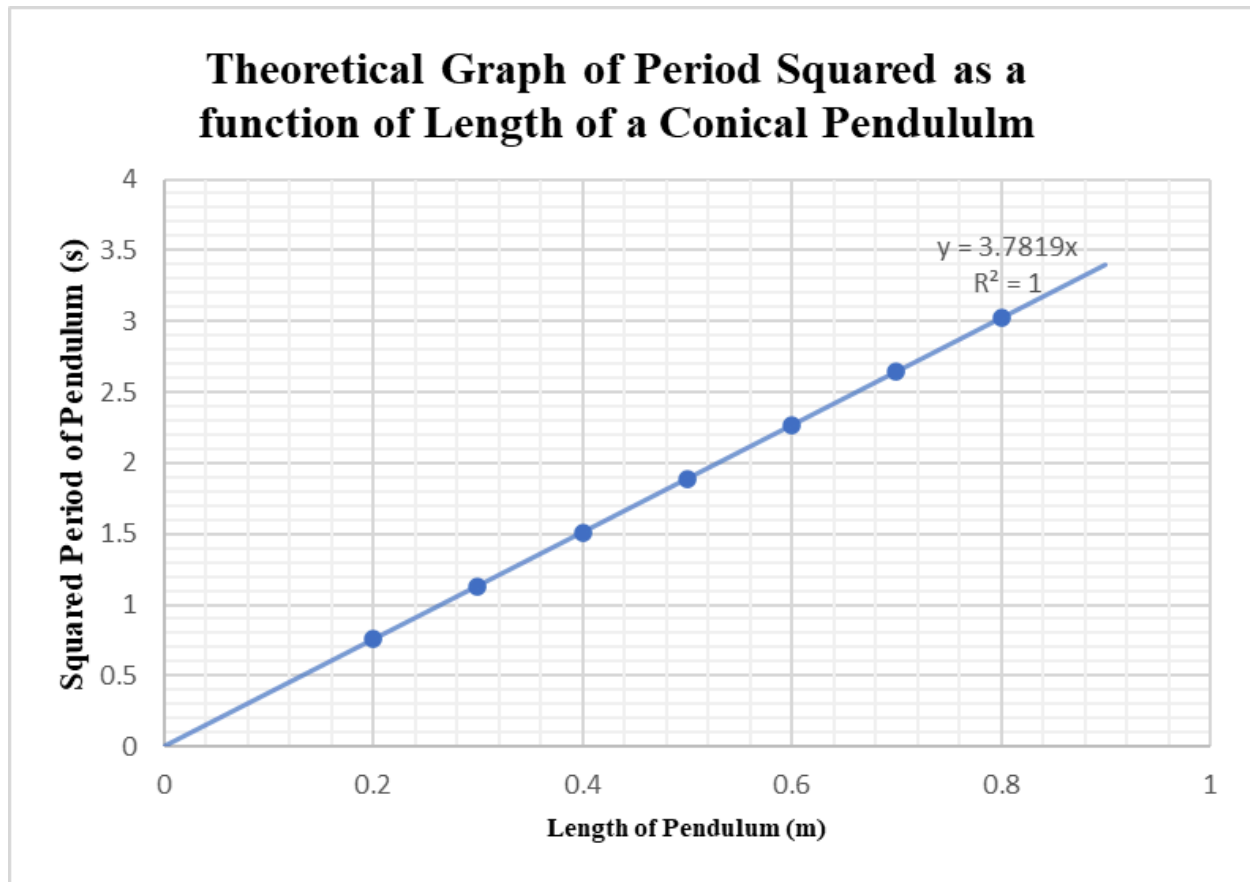
| Length (m) | Period (s) | Square of Period (s) |
|------------|------------|----------------------|
| 0.20       | 0.87       | 0.76                 |
| 0.40       | 1.23       | 1.51                 |
| 0.60       | 1.51       | 2.27                 |
| 0.80       | 1.74       | 3.03                 |

#### Sample Calculation for Squared Theoretical Period:

$$T = 2\pi\sqrt{\frac{\ell \cos \theta}{g}} = 2\pi\sqrt{\frac{0.20m \cos 20}{\frac{9.81m}{s^2}}} = 0.869667...s = 0.87s$$

$$(0.869667...s)^2 = 0.75632...s^2 = 0.76s^2$$

**Graph 3:** Curve Straightened Graph for Period as a Function of length on a Conical Pendulum using theoretical values obtained from the equation derived.



The slope of the trend line is 3.7819. We can see in the theoretical graph that as length increases, so does the square root of period. We can deem this relationship to be accurate because it complies with experimental results. As well, the graph intersects the y-axis at  $y = 0$  showing that with no length, there is no period.

#### Percent Error:

$$\text{Percent Error} = \left| \frac{\text{Theoretical} - \text{Experimental}}{\text{Theoretical}} \right| \cdot 100\%$$

$$= \frac{\frac{3.7819s^2}{m} - \frac{3.4436s^2}{m}}{\frac{3.7819s^2}{m}} \cdot 100\% = 8.945239...\% = 8.95\%$$

## Conclusion:

Changing the length of a conical pendulum has an observable effect on the period of its motion. This is demonstrated in the graphs created from my collected data. From the first graph, a square root relationship can be seen where as length increases, the period also increases but at an increasingly slower rate, resembling a square root relationship. This matches the proportionality predicted in the equation

$T = 2\pi\sqrt{\frac{\ell \cos \theta}{g}}$ .<sup>4</sup> The slope of the trend line on Graph 2,  $\frac{T^2}{\ell} = \frac{4\pi^2 \cos 20}{9.81}$  produced a constant value of 3.4436 ( $\pm 0.4036$ ) seconds squared per meter. This value is very reasonable because it is very close to the value for the constant obtained theoretically. Equation 3 can be used to calculate the theoretical period for each manipulated length. Then, period as a function of time can be graphed and then curve straightened. On the theoretical graph (Graph 3), the slope (a constant) is 3.7819 seconds squared per meter. The difference between the two values is very minimal at 0.3383 and produces a low percent error at 8.95%. This validates the experiment and confirms the relationship established earlier because both graphs resemble a square root function. I have a high degree of confidence in these results because of this.

Both the theoretical and the experimental results were curve straightened in order to clearly compare the two. In the original theoretical graph, you can see that a square root trend is followed. This is predicted correctly in my hypothesis because the period increases as the length does. The percent uncertainty is reasonable, sitting at approximately 12%. The line of best fit passed through all the error bars on my graphs which shows precision. In graph 2, there is a systematic offset error where the y-intercept of the graph is at 0.2586 rather than zero. This is a systematic error that was likely caused due to frictional effects and human error, where the timing was off due to reaction times and will be addressed in the evaluation.

Overall, I have a high degree of confidence in my results because my experimental results match scientific theory as predicted. There are no outliers in my data, however the precision could still be improved with more trials. This means my data values are of high quality and correctly reflect the theoretical relationship predicted in  $T = 2\pi\sqrt{\frac{\ell \cos \theta}{g}}$ .<sup>5</sup> As length increases, so does period, but not as fast as the length increases due to the square root proportionality.

## Evaluation:

*Systematic Errors:* These are errors that take away accuracy from my experiment.

1. A systematic error stands out the most when looking at my analysis. The zero offset error in Graph 2 is an error due to frictional effects and reaction time. Air resistance slows down the pendulum as it rotates, especially since the mass does not have a particularly aerodynamic shape. This would increase the period measurement for each trial, which is why the whole graph is offset higher than the predicted graph. Since mass will not affect the experiment as long as it is kept constant, an improvement would be to use a mass that is concentrated in a smaller space, or to use a smaller mass in general. A thinner string would also be advisable as the string I have used was

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<sup>4</sup> More, Hemant. (2020). *Conical Pendulum*. Retrieved February 16, 2024 from *The Fact Factor* <Conical Pendulum: Its time period, tension in string, numerical problems (thefactfactor.com)>

<sup>5</sup> Dean, Kevin. (2017.) *Conical Pendulum: Part 2 A Detailed Theoretical and Computational Analysis of Period, Tension and Centripetal Forces*.

rather thick and loose, meaning the string could trap a lot of air hereby slowing the pendulum and eventually leading to the decrease in angle of rotation.

2. I assumed the gravitational field strength is  $9.81\text{m/s}^2$  when it is really a little less than that in Calgary Alberta, where I conducted my experiment. An improvement is to make sure I use a more precise value of gravitational field strength proportionate to the altitude of my location when reproducing this experiment.
3. When the strings used on the pendulums were cut, they were measured correctly. However, I noticed that they stretched slightly when weighed down by the mass. This is a systematic error because the method of measuring and cutting the string is the same but they are weighed down by the same mass. This would cause the length measurements to be shorter than they actually are, which slightly affects the results. However, this error doesn't affect the data greatly when compared to other factors, such as air resistance as those would significantly reduce the angle, whereas this only adds a bit to the length of string.

*Random Errors:* These are errors that reduce precision in my experiment.

1. The biggest source of error would be the human error of reaction time. Although I personally timed each trial and kept the reaction time as constant as possible, I only started and stopped the timer after I was sure the bob passed through its starting point, which could have changed the measurement each trial. It would be better to set up a camera directly on top of the fixed point of rotation in order to use video analysis to time my trials. This could be done by taking taller retort stands and clamps to hold a camera in place. Then I could go back and analyze the frames per second for a more accurate period measurement.
2. I noticed on Graph 2 that the trend line just barely passed through the error bars on two of the points. This is because I have only conducted five trials for each length. Repeated trial error is random, and more trials should be conducted so that the effect of some inaccurate measurements are minimized as much as possible. Moreover, a wider range of data should be tested. Small lengths like 10 or 15 cm could be tested and, with a larger retort stand, values up to a meter should also be trialed. This would further confirm the square root relationship between length and period and increase precision of the data collected.

### **Extension:**

There are multiple types of pendulums that stem from a simple pendulum. The conical pendulum, for example, allows us to investigate concepts of circular motion and how period is affected by factors such as angle and length of pendulum. Double pendulums, connected pendulums, or twisting pendulums could be further investigated to explore torsion and effects of gravity and momentum.<sup>6</sup> However, a conical pendulum is comparably easier to recreate in a laboratory setting and provides us with valuable information on its motion.

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<sup>6</sup> Unknown (n.d.) Retrieved from *Physics LibreTexts* March, 2024. <15.5: Pendulums - Physics LibreTexts>

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4. Graphs Produced using Microsoft Excel.
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